PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Medical service utilization and direct medical cost of depression: a cross-sectional analysis of urban medical claims data from China
AUTHORS	Ding, Ruoxi; Zhu, Dawei; Wang, Yanshang; Yong, Ma; Shi, Xuefeng; He, Ping

VERSION 1 – REVIEW

REVIEWER	Lina Castro National Institute of Psychiatry “Ramón de la Fuente Muñiz”, Direction of Epidemiological and Psychosocial Research
REVIEW RETURNED	03-Nov-2021

GENERAL COMMENTS	I reviewed the manuscript ID bmjopen-2021-056422, entitled "Medical service utilization and direct medical cost of depression: Evidence from China urban medical claims data", although the topic is quite interesting, several sections of the manuscript are confusing and misleading several sections of the manuscript are confusing and misleading: 1.- line 7 of page 5, it seems that a piece of information is missing, please correct 2.- lines 55 to 58 page 5, transfer to RMB to US Dollar so that it is clearer to the reader. 3.- line 20 page 6, in order to clarify that it includes direct medical costs, for example: medications, hospitalization, diagnostic aids ..., in addition to out-of-pocket expenses; in other countries we do not know about China's medical services for depression. 4.- line 12 page 7, in China do you reimburse out-of-pocket expenses? 5.- line 16 page 7, what is the present value of the costs? 6 line 29 page 8 why do URBMI beneficiaries have higher medical and out-of-pocket costs? 7.- line 44 page 8, out-of-pocket expenses are a very low proportion (2.5%) of the median income, but on line 54 page 8, OOPs represent> 50% of outpatient care expenses and a third of care hospitable, is that correct? 8.- line 58 page 10, what do you mean by "the low treatment rate is explained by the low treatment cost" 9.- page 10, contrast your results with those of middle-income countries, for example Latin America. 10.- The values in the tables should be recorded in US dollars. Suggested: Cost analyzes of medical care for schizophrenia and

	depression in México, 2005-2013 doi: 10.1590 / 0102-311X00165816 On the other hand, the tables are incomplete and confusing… Thank you in advance

REVIEWER	Hannah König University Medical Center Hamburg-Eppendorf, Department of Health Economics and Health Services Research
REVIEW RETURNED	27-Jan-2022

GENERAL COMMENTS	The manuscript is well written and contains interesting information on health care utilization and costs for individuals with depression in China. I have some comments, which might help to improve the article: Methods • On page 4, line 55: I would suggest to state already at this point, what the ICD-10 diagnosis codes mean • The authors state that the patients with depression were identified between 2013 and 2016 in the dataset. Please clarify, how annual estimates were generated –were only the 12 months after the index date considered? • Page 5, line 14: I would recommend to convert all costs to a common, actual year, for example by using inflation rates from 2021 • I am missing a clear definition of the included cost categories. In the results section, outcomes for service providers like pharmacies and tertiary hospitals are reported, which have not been mentioned before. It would be interesting to get more details about the services usually provided by these institutions. • Furthermore, it is not clear to me why outpatient medication costs are not presented as a single cost category but ra-ther included in outpatient services. Maybe, this is a trait of the Chinese reimbursement system, which would also be interesting to read about in the introduction or methods section. • In general, if possible with the data used, it would be very interesting to estimate costs for a control group without depression and compare these findings. Results • In this study, out-of-pocket expenses have a high share in total costs. As this is not always the case in other countries, it would be helpful to get some more insights about the payment system in China (introduction section), and whether the estimates are reasonable/ to be expected (discussion section) • Table 4: An indicator what the coefficients show would be helpful. It could be interesting to see the differences in costs reported in US$/RMB • Table 4: What could be the reason why the number of outpatient visits are higher, but the costs were lower in women? Discussion • P. 9 lines 34-37: Please check if this comparison is reasonable, because, as mentioned above, in this study the outpatient medication costs were not yet reported separately, while this might have been the case in the other studies cited in the discussion • The limitations section in the discussion is rather short and I would suggest to discuss the advantages and disadvantes of this study more thoroughly. • A limitation of this study is that not all relevant cost categories were included due to the study data. Among others, depressive disorders

	cause high indirect costs that were not assessed in this study.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Dr. Lina Castro, National Institute of Psychiatry “Ramón de la Fuente Muñiz”
Comments to the Author:
I reviewed the manuscript ID bmjopen-2021-056422, entitled "Medical service utilization and direct medical cost of depression: Evidence from China urban medical claims data", although the topic is quite interesting, several sections of the manuscript are confusing and misleading several sections of the manuscript are confusing and misleading. Please see the attach file.
Response: We sincerely thank you for the valuable comments and suggestions, which helped us improved our manuscript. Below is the detailed response to each comment.

1.- line 7 of page 5, it seems that a piece of information is missing, please correct
Response: Thanks for your suggestion, we revised this part accordingly (Page 3, Line 27-32, highlighted, marked copy)

2.- lines 55 to 58 page 5, transfer to RMB to US Dollar so that it is clearer to the reader.
Response: Thanks for your suggestion. We revised these sentences accordingly (Page 4, Line 18-21, highlighted).

3.- line 20 page 6, in order to clarify that it includes direct medical costs, for example: medications, hospitalization, diagnostic aids ..., in addition to out-of-pocket expenses; in other countries we do not know about China's medical services for depression.
Response: Thanks for your comments. We revised the method part to explain and clarify the direct medical cost and out-of-pocket cost (Page 5, Line 48- 60, highlighted).

4.- line 12 page 7, in China do you reimburse out-of-pocket expenses?
Response: Thanks for your comments. In China, OOP cost cannot be reimbursed, and it was included as part of the direct medical cost. We added the explanation in the method part (Page 6, line4-10, highlighted).

5.- line 16 page 7, what is the present value of the costs?
Response: Thanks for your question. According the inflation rates of RMB reported by Statista (https://www.statista.com/statistics/270338/inflation-rate-in-china/), the present value of the costs was 1.103 times the cost we reported.

6 line 29 page 8 why do URBMI beneficiaries have higher medical and out-of-pocket costs?
Response: Thanks for your question. Compared with UEBMI beneficiaries, URBMI beneficiaries have lower reimbursement rate, and the proportion of severe cases was higher among URBMI patients who seek mental health care.

7.- line 44 page 8, out-of-pocket expenses are a very low proportion (2.5%) of the median income, but on line 54 page 8, OOPs represent> 50% of outpatient care expenses and a third of care hospitable, is that correct?
Response: Thanks for your question, annual average OOP cost was 786.4 RMB (not all the patients used inpatient services, so the relatively high OOP cost of inpatient care was balanced out by those only used outpatient services), which accounted for 2.5% of average disposable income of urban residents (31 440RMB). And average OOP cost per outpatient visit was 145.5 RMB, representing 40.5% of outpatient costs per visit. The average cost of inpatient care per admission was 9035.2 RMB, of which 2108.0 RMB (23.3%) was OOP cost. OOP cost was accounted for only a small proportion of annual disposable income, but represent a substantial proportion for outpatient or inpatient care cost. We hope this could answer your question.

8.- line 58 page 10, what do you mean by "the low treatment rate is explained by the low treatment cost"
Response: Thanks for your question. We discussed about the total direct medical cost for depression in urban China, which is only accounted for 0.04% of total health expenditure for urban residents in China. Because the treatment rate for depression in China was low, thus a large number of patients

with depression didn't use any treatment and didn't incur any expenditure for depression treatment. Therefore, the low treatment rate may explain the low total medical cost of depression. We further revised the description to clarify this issue (Page 9, Line59-60, highlighted).

9.- page 10, contrast your results with those of middle-income countries, for example Latin America.
Response: Thanks for your suggestion. We added the suggested literature as the reference to compare our result with middle-income countries in America (Page 10, Line 27, highlighted & reference 17).

10.- The values in the tables should be recorded in US dollars.
Suggested: Cost analyzes of medical care for schizophrenia and depression in México, 2005-2013 doi: 10.1590 / 0102-311X00165816 On the other hand, the tables are incomplete and confusing... Thank you in advance
Response: Thanks for your suggestion. We agree that using US dollars to show the cost will be easy to understand. However, the average exchange rate from 2013 to 2016 we used may result in potential bias and reduce the accuracy of our estimation, so we decided to use RMB in tables, but we added USD in all the cost provided in the manuscript. We further revised our tables.

Reviewer: 2
Mrs. Hannah König, University Medical Center Hamburg-Eppendorf
Comments to the Author:
The manuscript is well written and contains interesting information on health care utilization and costs for individuals with depression in China. I have some comments, which might help to improve the article:
Response: We sincerely thank you for the valuable suggestions, which helped us improved our manuscript, and the positive comments inspired us. Below is the detailed response to each comment.

Methods
• On page 4, line 55: I would suggest to state already at this point, what the ICD-10 diagnosis codes mean

Response: Thanks for your suggestion, we added the disease that the ICD-10 code represents (Page 5, Line 30-37, highlighted, marked copy).

• The authors state that the patients with depression were identified between 2013 and 2016 in the dataset. Please clarify, how annual estimates were generated –were only the 12 months after the index date considered?

Response: Thanks for your comments. The depression-related annual medical service utilization and cost was calculated based on calendar year, which means that the annual resource use and cost were the totals of depression-related service utilization and cost in a whole calendar year (2013, 2014, 2015 or 2016). We added this description in the method (Page 5, Line 48-51, highlighted)

• Page 5, line 14: I would recommend to convert all costs to a common, actual year, for example by using inflation rates from 2021

Response: Thanks for your suggestion. We considered to convert all costs to a common year using inflation rates. However, the converting may reduce the accuracy of our estimation, so we decided to provide the estimation of the cost incurred at the actual year (2013-2016), but we provide the inflation rate from 2016 to 2021 in the method part (Page 6, Line 9-11, highlighted).

• I am missing a clear definition of the included cost categories. In the results section, outcomes for

service providers like pharmacies and tertiary hospitals are reported, which have not been mentioned before. It would be interesting to get more details about the services usually provided by these institutions.

Response: Thanks for pointing this out. We added more details and the provider of outpatient and inpatient services in the method part (Page 6, line 5-9, highlighted).

• Furthermore, it is not clear to me why outpatient medication costs are not presented as a single cost category but rather included in outpatient services. Maybe, this is a trait of the Chinese reimbursement system, which would also be interesting to read about in the introduction or methods section.

Response: Thanks for your question. In China, outpatient medication was mainly prescribed by physicians during the outpatient visits, so the reimbursement system recorded medication cost as part of the cost of outpatient services.

• In general, if possible with the data used, it would be very interesting to estimate costs for a control group without depression and compare these findings.
Response: Thanks for this suggestion. We also considered to compare the cost between people with and without depression. Unfortunately, our study based on the claims data that extracted according to the ICD-10 code of depression, comparison of cost between people with depression and other groups is not applicable. We will conduct such analyses once we have the permission to use the claims data of population without depression.

Results
• In this study, out-of-pocket expenses have a high share in total costs. As this is not always the case in other countries, it would be helpful to get some more insights about the payment system in China (introduction section), and whether the estimates are reasonable/ to be expected (discussion section)

Response: Thanks for your question. We found 29% of the medical cost was paid by OOP payment for patient with depression, which was below the proportion of 30% for the universal access defined by the Organization for Economic Co-operation and Development (OECD), suggesting the proportion is reasonable. This finding was also consistent with the OOP rates of UEBMI and URBMI reported by previous studies. It has been suggested that the currently insurance scheme in urban China only pay for psychiatric treatment with medication, and support psychotherapy, community recovery services cannot be reimbursed, which may result in relatively higher proportion of OOP cost. We added discussion about this issue in the manuscript (Page 11, Line 3-11, highlighted).

• Table 4: An indicator what the coefficients show would be helpful. It could be interesting to see the differences in costs reported in US$/RMB

Response: Thanks for pointing this out, we added some explanation about the coefficients in the result part (Page 8, Line 23-30, highlighted).

• Table 4: What could be the reason why the number of outpatient visits are higher, but the costs were lower in women?

Response: Thanks for this excellent question. We also considered the reason about this issue. This result was consistent with studies on gender difference in mental health service use and cost from other countries. Female with depression utilized more mental health service compare with their male

counterparts, but they incurred lower cost. A possible explanation is that males are less likely to utilize health services until something severe happens. As a result, more extensive treatments are required, which in turn escalates health treatment costs. We also added this explanation in the discussion part (Page 12, Line 16-23, highlighted). We hope this could answer your question.

Discussion

• P. 9 lines 34-37: Please check if this comparison is reasonable, because, as mentioned above, in this study the outpatient medication costs were not yet reported separately, while this might have been the case in the other studies cited in the discussion

Response: Thanks for this valuable suggestion. We went through the research on cost of depression in developed countries that we cited (Germany and Japan). And it was true that medication costs were reported separately from the outpatient and inpatient medical costs. And both of these two studies additionally retrieved the cost of medication from other dataset to include it in the analysis, so we kept these comparisons. We really appreciate your advices.

• The limitations section in the discussion is rather short and I would suggest to discuss the advantages and disadvantes of this study more thoroughly.

Response: Thanks for your suggestion. We thoroughly revised the limitation section to thoroughly discuss the advantages and disadvantages of this study (Page 13, Line 11-21, highlighted).

• A limitation of this study is that not all relevant cost categories were included due to the study data. Among others, depressive disorders cause high indirect costs that were not assessed in this study.
Response: Thanks for this valuable suggestion. We added this in the limitation section (Page 13, Line 11-14, highlighted).

VERSION 2 – REVIEW

REVIEWER	Lina Castro National Institute of Psychiatry “Ramón de la Fuente Muñiz”, Direction of Epidemiological and Psychosocial Research
REVIEW RETURNED	07-May-2022

GENERAL COMMENTS	Reviewer: 1 I would like to thank the authors for the effort made to address my comments. I am pleased with the answer given to some of them, although I still have some major concerns about how the authors approached the reviewer’s suggestions concerning the medical services utilization costs comparison between inpatients and outpatients, and about some other points regarding the costs of depression and the interpretation of the results: Methods/Results What is the prevalence of depression by sex and age group in China? It is imperative that this is shown to interpret the results and compare them accurately. The 15-24 year age group incurs in OOP expenditure and is the most affected. What would this imply for the health system in China? How do the authors explain that patient with mild depression had the highest utilization of inpatient services if in multivariate analyses, patients with mild depression were associated with lower OOP expenditure? Is tertiary hospital utilization for healthcare of the patients with mental disorders in China, similar or does it contrast with the

	scientific literature? Line 39, Measures: spell out please “RMB” Line 44, Results: For instance, in the results, authors report “Predictors of medical service utilization and costs”. This is not completely accurate, since “Predictor” means “Characteristics associated”, this needs to be clearly stated. In Table 3, please specify that Severe, Moderate, Mild… correspond with levels of depression. Specify the meaning of “others” as well. In Table 4, annotate the currency for the estimated costs: RMD or USD? Line 28, spell out please “WHO” Line 25: “We found 29% of the medical cost was paid by OOP payment for patient with depression, which was below the proportion of 30% for the universal access defined by the Organization for Economic Co-operation and Development…” Please, clarify universal access to…? Furthermore, is necessary to complete the reference #22

REVIEWER	Hannah König University Medical Center Hamburg-Eppendorf, Department of Health Economics and Health Services Research
REVIEW RETURNED	23-May-2022

GENERAL COMMENTS	I have no further comments, but I would recommend English Proofreading.

VERSION 2 – AUTHOR RESPONSE

I would like to thank the authors for the effort made to address my comments. I am pleased with the answer given to some of them, although I still have some major concerns about how the authors approached the reviewer’s suggestions concerning the medical services utilization costs comparison between inpatients and outpatients, and about some other points regarding the costs of depression and the interpretation of the results:

Response: We sincerely thank you for the valuable suggestions, which helped us improved our manuscript, and the positive comments inspired us a lot. Below is the detailed response to each comment. Your generous works are highly appreciated.

Methods/Results

1. What is the prevalence of depression by sex and age group in China? It is imperative that this is shown to interpret the results and compare them accurately. The 15-24 year age group incurs in OOP expenditure and is the most affected. What would this imply for the health system in China?

Response: Thanks for the comments. We provide the estimated prevalence of treated depression by sex and age group in the result (Manuscript with tracked changes, Page 5, Line 41-44, highlighted). We also added explanation for the high OOP expenditure in 15-24 years and its implications in discussion (Page 10, Line 42-53, highlighted).

2. How do the authors explain that patient with mild depression had the highest utilization of inpatient services if in multivariate analyses, patients with mild depression were associated with lower OOP expenditure?

Response: Thanks for the question. We didn’t find reasonable explanation based on existing literature to illustrate the highest utilization of inpatient services among patients with mild depression. It is possible that people with mild depression may have more positive attitude toward mental health

service utilization in comparison to those with moderate or severe depression. But the higher inpatient service utilization may partly explain the lower OOP expenditure in people with mild depression since the reimbursement policies across cities in China were different. In some province, the medical expenditure could be reimbursed only for inpatient services.

3. Is tertiary hospital utilization for healthcare of the patients with mental disorders in China, similar or does it contrast with the scientific literature?

Response: Thanks for this question. Tertiary hospital utilization for healthcare of the patients with depression in China were similar with the scientific literature. This is in line with one study using electronic health record data in Shanghai, China, which compared mental health service utilization among inpatients from different types of health institution. Although integrating mental health services in community-based care has been implemented as the long-term goal, current lack of resources in primary health sector, combined with the low competitive ability in attracting mental health professionals to work in the community, resulted in the large gap in workforce and facilities with higher-tier hospitals. It has been reported that by 2015, less than 4% of inpatient beds for mental health services were allocated in primary care facilities in China.

Shi, J., et al., Disparities in mental health care utilization among inpatients in various types of health institutions: a cross-sectional study based on EHR data in Shanghai, China. BMC public health, 2019. 19(1):1-10.

Shi Chenhui, M.N., Wang Liying, Yi Lelai, Wang Bin, Study of the mental health resources in China. Chinese Journal of Health Policy, 2019. 12(2):1-7.

4. Line 39, Measures: spell out please “RMB”

Response: Thanks for the comments. We added the Renminbi for RMB in the manuscript (Page 4, Line 37, highlighted).

5. Line 44, Results: For instance, in the results, authors report “Predictors of medical service utilization and costs”. This is not completely accurate, since “Predictor” means “Characteristics associated”, this needs to be clearly stated.

Response: Thanks for the comments. We revised the statement in the result (Page 6, Line 57, highlighted).

6. In Table 3, please specify that Severe, Moderate, Mild... correspond with levels of depression. Specify the meaning of “others” as well.

Response: Thanks for the suggestion. We specified the different degree of depression in the method (Page 4, Line 48-60, highlighted).

7. In Table 4, annotate the currency for the estimated costs: RMD or USD?

Response: Thanks, we annotated the currency in Table 4.

8. Line 28, spell out please “WHO”

Response: Thanks for the comments. We added the World Health Organization for WHO in the manuscript (Page 8, Line 41, highlighted).

9. Line 25: “We found 29% of the medical cost was paid by OOP payment for patient with depression, which was below the proportion of 30% for the universal access defined by the Organization for Economic Co-operation and Development...”

Please, clarify universal access to...?

Furthermore, is necessary to complete the reference #22

Response: Thanks for the comments. We clarified this issue in the discussion and complete the reference 22 (Page 9, Line 39-342, highlighted).

VERSION 3 – REVIEW

REVIEWER	Lina Castro National Institute of Psychiatry “Ramón de la Fuente Muñiz”, Direction of Epidemiological and Psychosocial Research
REVIEW RETURNED	12-Jul-2022

GENERAL COMMENTS	Dear Editor BMJ Open Thank you for this opportunity to review again the manuscript ID: bmjopen-2021-056422.R2 I appreciate the effort made by the authors for the changes in their document incorporating my suggestions. I think these changes have improved the manuscript. Thanks to the authors for this new version.